

Networking Basics — Detailed Lecture Notes

JKSSB Computer Awareness

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40.1 What a network is: nodes, hosts, clients, servers

A computer network is formed when two or more computers and related devices are connected so that they can share data and resources. The shared resources include files, printers, an internet connection, and application software. On the office desktop, opening a shared folder, printing to a shared printer, and browsing the web all use the same underlying network.

Definition (Computer network). *A set of two or more connected computing devices (nodes) that can exchange data and share resources such as files, printers, and an internet connection.*

Every participant in a network is called a **node**. A node can be a computer, a printer, a switch, or any device with a network address. A **host** is a node that can send or receive data over the network, such as a desktop computer, a laptop, or a network printer. In practice every host is a node, but the word host is used when the device is an end point that produces or consumes data.

Term	Meaning in this lesson
Network	Two or more connected devices that share data and resources.
Node	Any participant device in the network.
Host	A node (computer, printer) that sends or receives data.
Client	The device or program that requests a service.
Server	The computer that provides the service and answers requests.

The client–server pair is the most examined relationship in this block. The **client** is the device (or program) that requests a service — a web page, a file, or a print job. The **server** is the computer that provides that service: it stores the shared files or pages and answers the requests of many clients. When the office desktop opens a file stored on the department’s shared machine, the desktop is the client and the shared machine is the server.

Definition (Client and server). *The client requests a service; the server stores the shared resource and provides it in response to client requests.*

40.2 Networks by area: PAN, LAN, MAN, WAN

Networks are classified by the geographical area they cover. Four names must be learned together with their full forms, because the exam tests both the expansion and the coverage in the same item.

Type	Full form	Coverage and example
PAN	Personal Area Network	Around one person: phone, laptop, and ear-phones linked by Bluetooth.
LAN	Local Area Network	One building or campus: an office, school, or laboratory network.
MAN	Metropolitan Area Network	One city: a bank's branches linked across a town.
WAN	Wide Area Network	A country or the world: the Internet is the largest WAN.

A **PAN** (Personal Area Network) covers the smallest area — the devices around a single person, such as a phone connected to a laptop and wireless earphones. A **LAN** (Local Area Network) covers one building or campus, such as all the computers of one office floor connected through a switch. A **MAN** (Metropolitan Area Network) spans a city, for example the branches of one organisation linked across a town. A **WAN** (Wide Area Network) spans a country or the whole world; the Internet is the largest example of a WAN.

Definition (PAN, LAN, MAN, WAN). *PAN covers one person; LAN covers one building or campus; MAN covers one city; WAN covers a country or the world.*

Example 40.2.1. *A question gives four coverages — one person, one building, one city, and the whole world — and asks for the matching names. Read the area first: person means PAN, building means LAN, city means MAN, and worldwide means WAN. The area decides the answer before the full form is even expanded.*

40.3 WLAN, intranet, Internet, and ISP

Three further names complete the vocabulary. A **WLAN** (Wireless Local Area Network) is a LAN built with wireless signals instead of cables: the office Wi-Fi through which laptops connect without a cable is a WLAN. An **intranet** is a private network inside one organisation, usable only by its own staff — for example the company's internal portal. The **Internet** is the worldwide public network of connected networks, open to anyone with access. The two words differ by one prefix and are a standing trap: intranet is private to one organisation, while the Internet is public and worldwide.

An **ISP** (Internet Service Provider) is the company that provides a customer with access to the Internet. The home or office connects through its ISP — over a phone line, cable, fibre, or mobile link — and the ISP carries the traffic onward to the Internet.

Definition (WLAN, intranet, Internet, ISP). *A WLAN is a wireless LAN; an intranet is a private organisational network; the Internet is the worldwide public network; an ISP (Internet Service Provider) gives customers access to the Internet.*

40.4 Topology: bus, star, ring, mesh, tree

The word **topology** means the arrangement by which nodes are connected in a network. Five topologies are examined: bus, star, ring, mesh, and tree. Each one decides how data travels and what happens when one cable or device fails, and the exam repeatedly asks exactly that: which topology has a central device, and which one fails fully when the main cable breaks.

In a **bus** topology, all nodes share one single main cable called the bus. It is simple and cheap for a small network. Data sent by one node travels along the whole bus and is accepted by the addressed node. Its weakness follows from its shape: if the main cable breaks, the whole network fails.

In a **star** topology, every node connects to one central device (a hub or a switch), and all data passes through that centre. If one spoke cable fails, only that node is affected and the rest keep working. But if the central device itself fails, the whole network stops. The ordinary office LAN — one switch with many desktops plugged into it — is a star, and it is the most common topology in exam examples.

In a **ring** topology, each node connects to exactly two neighbours, forming a closed circle. Data travels around the ring, usually in one direction, passed from node to node until it reaches its destination. Each intermediate node repeats the data it is not addressed to. A break at one point disturbs the whole ring, unless a second (dual) ring provides a backup path.

In a **mesh** topology, every node has a direct link to many (in a full mesh, all) other nodes. It is the most reliable topology: if one path fails, another path carries the data. It is also the costliest, because it needs the most cable and the most ports.

In a **tree** topology, several star networks are joined through a central backbone, like the branches of a tree. It suits a large campus: each department is a star, and the stars join at the centre. Its weakness is the backbone: if the backbone fails, whole branches lose connection.

Topology	Shape	What one failure does
Bus	One shared cable	Main cable breaks: the whole network fails.
Star	Centre with spokes	One spoke fails: one node affected; centre fails: all fail.
Ring	Closed circle	One break disturbs the ring.
Mesh	Many direct links	One path fails: traffic uses another path.
Tree	Stars joined on a backbone	Backbone fails: whole branches fail.

Definition (Topology). *The physical or logical arrangement of nodes in a network: bus (one cable), star (central device), ring (closed circle), mesh (many direct links), and tree (stars joined on a backbone).*

Example 40.4.1. *An item states, “In which topology does the failure of the central device bring down the whole network?” Trace the table: only the star routes every message through one centre, so the answer is star. If the stem instead says the main cable broke and everything stopped, the shape is the single shared cable, so the answer is bus.*

40.5 Transmission media: guided and unguided

Transmission media are the paths that carry data signals from one device to another. They divide into two families. **Guided (wired) media** confine the signal inside a physical cable. **Unguided (wireless) media** carry the signal through air or space without a cable.

There are three guided media. **Twisted-pair cable** consists of pairs of copper wires twisted together; it is the ordinary telephone and office-LAN cable — cheap and suitable for short ranges. **Coaxial cable** has a central copper wire with insulation and a metal shield around it; it is used for cable TV and older LANs and has better shielding than twisted pair. **Fibre-optic cable** uses thin glass or plastic strands carrying light signals; it is the fastest medium with the longest range and

is immune to electrical noise, but it is also the costliest. Ranked by speed and range, fibre-optic comes first, then coaxial, then twisted pair.

There are four unguided media at this level. **Radio waves** travel in all directions and are used for FM radio and for short-range WLAN and Bluetooth links. **Microwaves** travel in a straight line and need line of sight; they are used for point-to-point links between towers. **Satellite links** are microwave signals relayed through a satellite; they cover very long distances but introduce delay. **Infrared** works over a very short range, is blocked by walls, and is used for TV remotes.

Definition (Guided and unguided media). *Guided media carry signals inside a cable (twisted pair, coaxial, fibre-optic); unguided media carry signals through air or space (radio waves, microwaves, satellite links, infrared).*

40.6 Networking devices and their roles

Ten devices are examined, and device–role matching is the highest-yield item type of this lesson. Each device has one role to memorise.

A **NIC** (Network Interface Card) is the card — or the built-in circuit on the motherboard — that lets a computer physically connect to a network. Without it the machine has no network port to plug into.

A **modem** (modulator-demodulator) converts digital computer signals into analogue signals for a telephone line and back again. It is the device that connects a home or office to the ISP over a line that cannot carry digital signals directly.

A **hub** connects several nodes in a star and broadcasts every incoming signal to all of its ports; the addressed node accepts the data and the rest ignore it. A **switch** also connects nodes in a star, but it forwards each message only to the addressed port. The switch is therefore smarter and faster than the hub, because it does not flood every port with every message.

A **repeater** regenerates a weak signal so that it can travel further along a long cable. It works at the signal level only: it does not choose a path and does not filter traffic. A **bridge** connects two LAN segments and filters traffic, passing across only the data meant for the other side.

A **router** connects different networks — for example a home LAN to the ISP’s network — and chooses the best path for each packet of data. A **gateway** connects networks that use different protocols or data formats, translating between them so the two sides can understand each other. An **access point** is the wireless base station that lets Wi-Fi devices join a wired network. A **firewall** blocks unauthorised access while letting legitimate traffic pass; it is a security device that can be hardware, software, or both.

Device	Role (one line)
NIC (Network Interface Card)	Connects a computer to the network.
Modem	Converts digital to analogue and back; links the site to the ISP.
Hub	Broadcasts every incoming signal to all ports.
Switch	Forwards each message only to the addressed port.
Repeater	Regenerates a weak signal for a longer run.
Bridge	Joins two LAN segments and filters traffic between them.
Router	Joins different networks and chooses the best path.
Gateway	Translates between networks using different protocols.
Access point	Lets wireless (Wi-Fi) devices join the wired network.
Firewall	Blocks unauthorised access; a security device.

Definition (The four most confused pairs). *Hub sends to all, switch sends to the addressed port; repeater regenerates the signal, router chooses the path; bridge joins two similar LAN segments, gateway translates between different protocols; modem connects to the line, firewall guards the entry.*

40.7 Exam retrieval and common confusions

Six distinctions prevent most errors on this topic.

1. The client **requests** and the server **provides**. Do not swap the roles.
2. Read the area first: one person is PAN, one building is LAN, one city is MAN, worldwide is WAN.
3. **Intranet** is private to one organisation; the **Internet** is public and worldwide.
4. A hub broadcasts to **all** ports; a switch forwards only to the **addressed** port.
5. A repeater only **regenerates** the signal; the router **chooses the path**. A bridge joins two LAN segments; a gateway translates between **different** protocols.
6. Fibre-optic is the **fastest** guided medium; twisted pair is the slowest of the three cables.

Example 40.7.1. *An item asks which device “connects different networks and selects the best path for each packet.” Walk the device table: the hub and switch stay inside one network, the repeater only strengthens the signal, and the gateway translates protocols — only the router joins different networks and chooses paths, so the answer is router.*

Exercises

1. Distinguish a node, a host, a client, and a server.
2. Expand PAN, LAN, MAN, WAN, WLAN, NIC, and ISP, and state the coverage of each network type.
3. Distinguish an intranet from the Internet.
4. Describe the five topologies and state, for each, what a single failure does.
5. Distinguish guided from unguided transmission media, giving the three cables and the four wireless media of this lesson.
6. Match each of the ten devices of this lesson to its one-line role.

Solutions

1. A node is any participant device in the network. A host is a node (computer, printer) that sends or receives data. A client is the device or program that requests a service; a server is the computer that provides the service and answers client requests.
2. PAN — Personal Area Network (around one person); LAN — Local Area Network (one building or campus); MAN — Metropolitan Area Network (one city); WAN — Wide Area Network (a country or the world); WLAN — Wireless Local Area Network (a LAN built with wireless signals); NIC — Network Interface Card (lets a computer connect to a network); ISP — Internet Service Provider (gives customers Internet access).

3. An intranet is a private network inside one organisation, usable only by its staff. The Internet is the worldwide public network of connected networks, open to anyone with access.
4. Bus: one shared cable; a break in the main cable fails the whole network. Star: every node joins one centre; a spoke failure affects one node, a centre failure stops all. Ring: closed circle of neighbours; one break disturbs the ring. Mesh: many direct links; one failed path leaves others. Tree: stars joined on a backbone; a backbone failure drops whole branches.
5. Guided media confine the signal in a cable: twisted-pair (cheap, short range), coaxial (shielded, e.g. cable TV), and fibre-optic (light signals; fastest, longest range, costliest). Unguided media use air or space: radio waves (all directions, short range), microwaves (straight line, towers), satellite links (long distance, with delay), and infrared (very short range, blocked by walls).
6. NIC connects the computer; modem converts digital–analogue for the line to the ISP; hub broadcasts to all ports; switch forwards to the addressed port; repeater regenerates a weak signal; bridge joins two LAN segments with filtering; router joins different networks and chooses the path; gateway translates between different protocols; access point admits Wi-Fi devices; firewall blocks unauthorised access.